COMPONENT INCLUSIONS INTO THE SIMPLICIAL TOTAL SITE ARE CONTINUOUS AND COCONTINUOUS, WITH EXPLICIT SHRIEK FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the single omission (source line 1056, 'we omit the arguments showing these two constructions give mutually inverse maps') is the inverse check of the adjunction bijection, auxiliary to parts (3) and (6) which are constructed in full; the author's typo 'ammner' (source line 1066, for 'manner') is preserved verbatim; the left adjoints g_{n!} established here, with their component formula, are used by card L46148 of this batch.

Editorial note. The label lemma-restriction-to-components-site names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Morphisms of topoi from cocontinuous functors. Sites chapter, \label{sites-lemma-cocontinuous-morphism-topoi}. Line 1045: produces the morphism of topoi g_n and part (1) in both cases A and B.
- Existence of the shriek functor for a cocontinuous functor. Sites chapter, \label{sites-lemma-when-shriek}. Line 1046: existence of the sheaf-level left adjoint g^{Sh}_{n!} (part (2)).
- The shriek adjoint on abelian sheaves. Modules on Sites chapter, \label{sites-modules-lemma-g-shriek-adjoint}. Line 1048: existence of the abelian left adjoint g_{n!} (part (5)).
- Coverings and fibre products in the simplicial total site. Same chapter, \label{lemma-simplicial-site-site} and \label{lemma-simplicial-cocontinuous-site}. Lines 995-1041: in case A resp. case B, images of coverings are coverings by definition and fibre products computed in C_n agree with those in C_total (the claim in the cited lemma's proof).

Source reference. Original reference 09WD: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 844–844. Prerequisite pointer invoked at proof

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line(s) 1009; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WC: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 797–797. Prerequisite pointer invoked at proof line(s) 996; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00XO: sources/stacks-chow/tags.txt, source lines 1226–1226. Prerequisite pointer invoked at proof line(s) 1022; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00XR: sources/stacks-chow/tags.txt, source lines 1229–1229. Prerequisite pointer invoked at proof line(s) 1023; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04BG: sources/stacks-chow/tags.txt, source lines 5651–5651. Prerequisite pointer invoked at proof line(s) 1025; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 971; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE the functor $\mathcal{C}_n \to \mathcal{C}_{total}$, $U \mapsto U$ is continuous and cocontinuous. The associated morphism of topoi $g_n : Sh(\mathcal{C}_n) \to Sh(\mathcal{C}_{total})$ satisfies*

1. *g_n^{-1} associates to the sheaf $\mathcal{F}$ on $\mathcal{C}_{total}$ the sheaf $\mathcal{F}_n$ on $\mathcal{C}_n$,*
2. *$g_n^{-1} : Sh(\mathcal{C}_{total}) \to Sh(\mathcal{C}_n)$ has a left adjoint $g_{n!}^{Sh}$,*
3. *for $\mathcal{G}$ in $Sh(\mathcal{C}_n)$ the restriction of $g_{n!}^{Sh}\mathcal{G}$ to $\mathcal{C}_m$ is $\coprod_{\varphi:[n]\to[m]} f_\varphi^{-1}\mathcal{G}$,*
4. *$g_{n!}^{Sh}$ commutes with finite connected limits,*
5. *$g_n^{-1} : Ab(\mathcal{C}_{total}) \to Ab(\mathcal{C}_n)$ has a left adjoint $g_{n!}$,*
6. *for $\mathcal{G}$ in $Ab(\mathcal{C}_n)$ the restriction of $g_{n!}\mathcal{G}$ to $\mathcal{C}_m$ is $\bigoplus_{\varphi:[n]\to[m]} f_\varphi^{-1}\mathcal{G}$, and*
7. *$g_{n!}$ is exact.*

Proof. Case A. If $\{U_i \to U\}_{i\in I}$ is a covering in $\mathcal{C}_n$ then the image $\{U_i \to U\}_{i\in I}$ is a covering in $\mathcal{C}_{total}$ by definition (Lemma 09WC). For a morphism $V \to U$ of $\mathcal{C}_n$, the fibre product $V \times_U U_i$ in $\mathcal{C}_n$ is also the fibre product in $\mathcal{C}_{total}$ (by the claim in the proof of Lemma 09WC). Therefore our functor is continuous. On the other hand, our functor defines a bijection between coverings of U in $\mathcal{C}_n$ and coverings of U in $\mathcal{C}_{total}$. Therefore it is certainly the case that our functor is cocontinuous.

Case B. If $\{U_i \to U\}_{i\in I}$ is a covering in $\mathcal{C}_n$ then the image $\{U_i \to U\}_{i\in I}$ is a covering in $\mathcal{C}_{total}$ by definition (Lemma 09WD). For a morphism $V \to U$ of $\mathcal{C}_n$, the fibre product $V \times_U U_i$ in $\mathcal{C}_n$ is also the fibre product in $\mathcal{C}_{total}$ (by the claim in the proof of Lemma 09WD). Therefore our functor is continuous. On the other hand,

our functor defines a bijection between coverings of U in $\mathcal{C}_n$ and coverings of U in $\mathcal{C}_{total}$. Therefore it is certainly the case that our functor is cocontinuous.

At this point part (1) and the existence of $g_{n!}^{Sh}$ and $g_{n!}$ in cases A and B follows from Sites, Lemmas 00XO and 00XR and Modules on Sites, Lemma 04BG.

Proof of (3). Let $\mathcal{G}$ be a sheaf on $\mathcal{C}_n$. Consider the sheaf $\mathcal{H}$ on $\mathcal{C}_{total}$ whose degree m part is the sheaf

$$\mathcal{H}_m = \coprod_{\varphi:[n]\to[m]} f_\varphi^{-1}\mathcal{G}$$

given in part (3) of the statement of the lemma. Given a map $\psi : [m] \to [m']$ the map $\mathcal{H}(\psi) : f_\psi^{-1}\mathcal{H}_m \to \mathcal{H}_{m'}$ is given on components by the identifications

$$f_\psi^{-1} f_\varphi^{-1}\mathcal{G} \to f_{\psi\circ\varphi}^{-1}\mathcal{G}$$

Observe that given a map $\alpha : \mathcal{H} \to \mathcal{F}$ of sheaves on $\mathcal{C}_{total}$ we obtain a map $\mathcal{G} \to \mathcal{F}_n$ corresponding to the restriction of α_n to the component $\mathcal{G}$ in $\mathcal{H}_n$. Conversely, given a map $\beta : \mathcal{G} \to \mathcal{F}_n$ of sheaves on $\mathcal{C}_n$ we can define $\alpha : \mathcal{H} \to \mathcal{F}$ by letting α_m be the map which on components

$$f_\varphi^{-1}\mathcal{G} \to \mathcal{F}_m$$

uses the maps adjoint to $\mathcal{F}(\varphi) \circ f_\varphi^{-1}\beta$. We omit the arguments showing these two constructions give mutually inverse maps

$$\mathrm{Mor}_{Sh(\mathcal{C}_n)}(\mathcal{G}, \mathcal{F}_n) = \mathrm{Mor}_{Sh(\mathcal{C}_{total})}(\mathcal{H}, \mathcal{F})$$

Thus $\mathcal{H} = g_{n!}^{Sh}\mathcal{G}$ as desired.

Proof of (4). If $\mathcal{G}$ is an abelian sheaf on $\mathcal{C}_n$, then we proceed in exactly the same ammner as above, except that we define $\mathcal{H}$ is the abelian sheaf on $\mathcal{C}_{total}$ whose degree m part is the sheaf

$$\bigoplus_{\varphi:[n]\to[m]} f_\varphi^{-1}\mathcal{G}$$

with transition maps defined exactly as above. The bijection

$$\mathrm{Mor}_{Ab(\mathcal{C}_n)}(\mathcal{G}, \mathcal{F}_n) = \mathrm{Mor}_{Ab(\mathcal{C}_{total})}(\mathcal{H}, \mathcal{F})$$

is proved exactly as above. Thus $\mathcal{H} = g_{n!}\mathcal{G}$ as desired.

The exactness properties of $g_{n!}^{Sh}$ and $g_{n!}$ follow from formulas given for these functors. □